

# **Chapter 12**

## **System Implementation 2**

**(Training and System Changeover)**

# 3. Training

# Training



- **Who to be trained** – users, operations personnel, technical personnel and top management.
- **Purpose** – to impart knowledge and skills related to the new system and implementation.
- **Amount of training** - depend upon the complexity of the system and the skills presently available (training gap).
- **Responsibilities of training** – Project manager and System Analyst.



# Purposes of Training

- **Transfer of knowledge and skills.**  
Familiarises users knowledge and skills in tasks performance and systems functions.
- **Facilitates User Involvement / Commitment.**  
Training is the best way to get users involved with the new system and commitment.
- **Offers Opportunities to Spot and Rectify Errors on the System.** Offer suggestions for rectification of the errors.

# Methods of Training



# 1. Instructor-Based

- **Meaning** - involving an instructor who will explain and demonstrate to the participants the skills and procedures
- **Options available** – public courses, tailored in-house (external trainers), tailored- in-house (internal trainers)
- **Advantages** - easy to develop and conduct, trainer as subject expert.
- **Disadvantages** - difficulties in logistic arrangement, higher cost for frequent training sessions.



# External Training (Public Course)

- **Provider** - Many hardware and software **vendors** offer training programs free or at a nominal cost for the products they sell to their customers.
- **Trainer** – Vendor's experienced trainers
- **Venue** – Vendor's site. A standard training for all clients as a public course.
- **Mode** – Depending on the nature and needs of the training course.
- **Logistics arrangement** – By vendor eg. equipment set-up, software installations, booking of rooms, contact trainers and trainees etc

# Internal Training (Internal Trainers)

- **Provider** – Some organizations would prefer to conduct their own training
- **Trainer** – Company's own expert trainers
- **Venue** – Company's choice: internal (eg. lab) or external training room
- **Mode** – Depending on the nature and needs of the training course.
- **Logistics Arrangement** – All logistics arrangement are responsibilities of the company.

# Internal Training (External Trainer)

- **Provider** - If a large number of people need training, **company** might prefer to arrange its own training classes.
- **Trainer** – Vendor's trainer
- **Venue** – Client's own premises or preferred location.
- **Mode** – Depending of the nature and needs of the training course.
- **Logistics Arrangement** – Major part by the company, minor part by the external trainer.

# Advantages of Internal Training

- **Customisation.** Trainer would be able to tailor the training to suit the company's training needs.
- **Flexible.** Conduct of training classes can also be more flexible in terms of time, pace, mode and venue.
- **Control.** Better control over time, costs and quality of training.
- **Assessment.** Assessment on the effectiveness of training can also be designed and conducted after training.

# Disadvantages of Internal Training

- **Logistics Arrangement.** The client/company would need to make all the necessary logistics arrangement.
- **Exchange experiences.** No opportunity to exchange experiences and knowledge with staff from other companies.
- **Qualification.** The internal trainer may not be qualified.

## 2. Computer-Based (CBT)

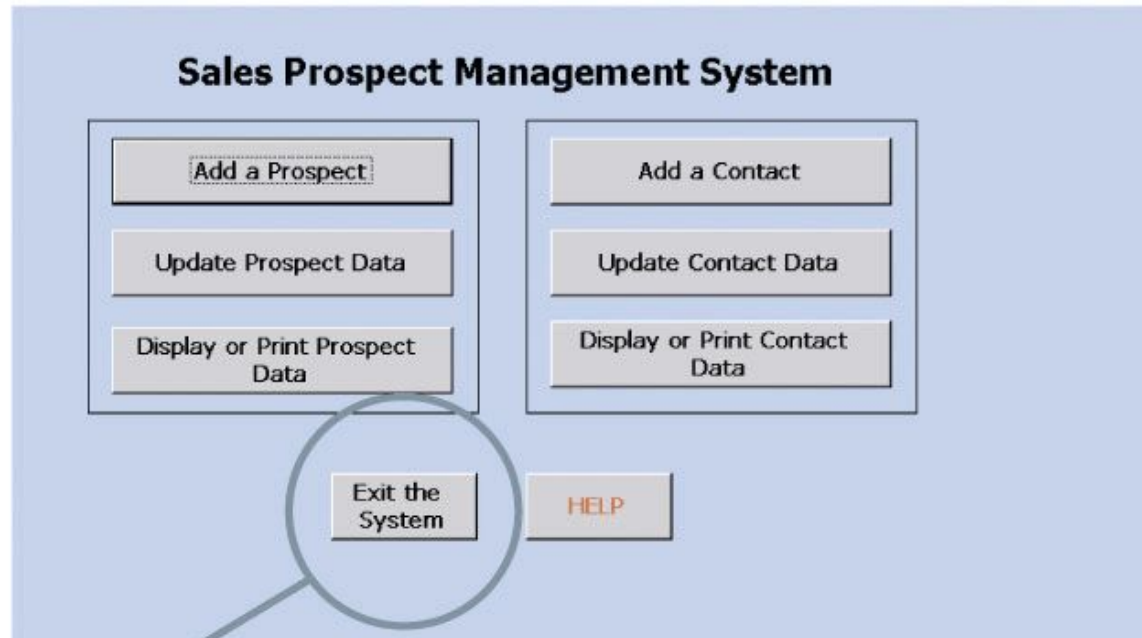
- **Provider** – The software will in most cases, purchased from a vendor.
- **Mode** – Use of software to provide training on-demand eg PC, online, website.
- **Advantages** :
  - can be used as and when training is required (flexible time and pace)
  - lower on-going training costs (no trainers)
  - Flexible training time and pace
  - able to track what the participants have learnt
- **Disadvantages** - high initial investment (if developed), complex to develop, training my



# CBT – Example 1

## Sales Prospect Management System Lesson 1: Entering and Exiting the System

- Start your computer. When the desktop appears, double-click the Sales Prospect Management System icon. Your screen should display the main switchboard shown below.



- To exit the Sales Prospect Management System, click the Exit the System button.

# 3. Self-Training

- **Meaning** - Instructor / User manuals are prepared and given to users to pick up the skills and knowledge.
- **Advantages** - manuals are easy to develop, low initial investment (assuming no software is involved) and less logistic requirements.
- **Disadvantages** – not effective, difficult to control the completion of the training.

# Self Training - Podcast and Webinars

- Podcast

- **Meaning** : Web-based *broadcast* that allows a user to download multimedia or simple **audio** files.
- **Software requirements** : a music player software eg. Microsoft Media Player, iTunes®, other MP3 player.
- **Time** : downloaded as and when required or scheduled
- **Advantages** - can play using any portable devices eg. PDAs, iPods, and even handphones at any place and at any time.

- Webinars

- **Acronym** : *Web* and *seminar*
- **Web-based** training approaches : 2 types
  - Interactive training session with instructor via internet.
  - Non-interactive, pre-recorded Webinar session can also be made available to the participants.
- **Advantage** - user can 'attend' training whenever he or she needs it

# Example of Podcast

HOME | DEVOTIONALS | RADIO & TV | BIBLE STUDY | BLOG | STORE | ABOUT RBC  
From the Publishers of "Our Daily Bread"



## words to live by

Radio

HOW WE ARE FUNDED | HOW TO DONATE

RBC INTERNATIONAL

**INSIDE WORDS TO LIVE BY**  
[HOME](#)  
[STATIONS LISTING](#)  
[HOST BIOS](#)  
[SHARE YOUR STORY](#)  
[HISTORY](#)  
[PERSONAL RELATIONSHIP WITH GOD](#)  
[PAST PROGRAMS](#)

**WORDS TO LIVE BY**  
**TOOLS:**  
**TEXT SIZE:**   
**ORIGINALLY AIRED ON: FRIDAY, JULY 25, 2008**  
**KATHY'S STORY - DEPRESSION, SALVATION**  
[Listen Now](#) | [Download](#) | [Podcast](#)  
  
**JOHN 17:3; PHILIPPIANS 4:6; JOHN 8:31-32**  
Kathy grew up in Hungary, a country brutalized under Communist rule. After fleeing to the United States in 1956, she found freedom from oppression. But it wasn't long before Kathy fell victim to another form of captivity—addiction. In this inspiring edition of "Words To Live By," you'll hear about the true Source of freedom.

**WORDS TO LIVE BY ARCHIVE**  

|    |           |    |    |    |    |    |  |   |
|----|-----------|----|----|----|----|----|--|---|
| <  | JULY 2008 |    |    |    |    |    |  | > |
| S  | M         | T  | W  | T  | F  | S  |  |   |
| 29 | 30        | 1  | 2  | 3  | 4  | 5  |  |   |
| 6  | 7         | 8  | 9  | 10 | 11 | 12 |  |   |
| 13 | 14        | 15 | 16 | 17 | 18 | 19 |  |   |
| 20 | 21        | 22 | 23 | 24 | 25 | 26 |  |   |
| 27 | 28        | 29 | 30 | 31 | 1  | 2  |  |   |

**SEARCH**  
☐ RBC MINISTRIES  
☒ WORDS TO LIVE BY  
  
**ADVANCED SEARCH**  
  
**RELATED RESOURCES**  
**OUR DAILY BREAD**  

- [Juneteenth](#)
- [Riding A Rollercoaster](#)

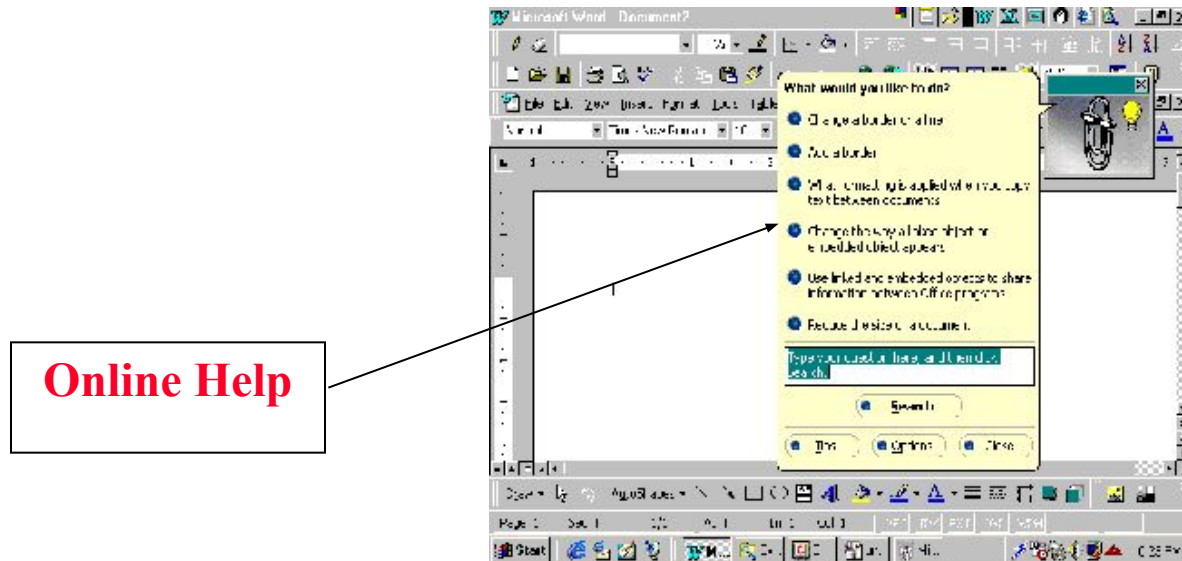
[More >>](#)  
**MY UTMOST FOR HIS HIGHEST**  

- ["By the Grace of God I Am What I Am"](#)
- [The Riches of the Destitute](#)

[More >>](#)  
**DISCOVERY SERIES**

# On-line Help/Tutorial

- Enables users to get help while performing the job.
- **Advantage** - it provides on-demand training
- **Disadvantage** - high initial investment and complex to develop



# Case Studies and Exercises

- **Meaning** - both classroom tuition and CBT will be supplemented with case studies and exercise
- **Advantages** – simulate real environment, easy to produce and they do help the user understand exactly how the new software will work.

# Deliver Effective Training - Steps

1. **Training objective.** Establish what an organisation wants the training to achieve.
2. **Training needs analysis.** Identify gap between what employees know now and what they must know to perform their job effectively.
3. **Course design.** Covers duration, course content, trainers.
4. **Conduct training.** The training is implemented using appropriate methodology eg. instructor-based. All logistic arrangements have to be made.
5. **Assessment.** After the learning has taken place it must be assessed to see if the training is effective.

# Exercise for Students

**MEX Ltd has just purchased a new online attendance system which uses the RFID (Radio Frequency Identification).**

## **Questions :**

- 1. Suggest one approach to training of all the staff in MEX Ltd.**
- 2. Give reasons for your choice.**

# Suggested Answer

## Suggested training method:

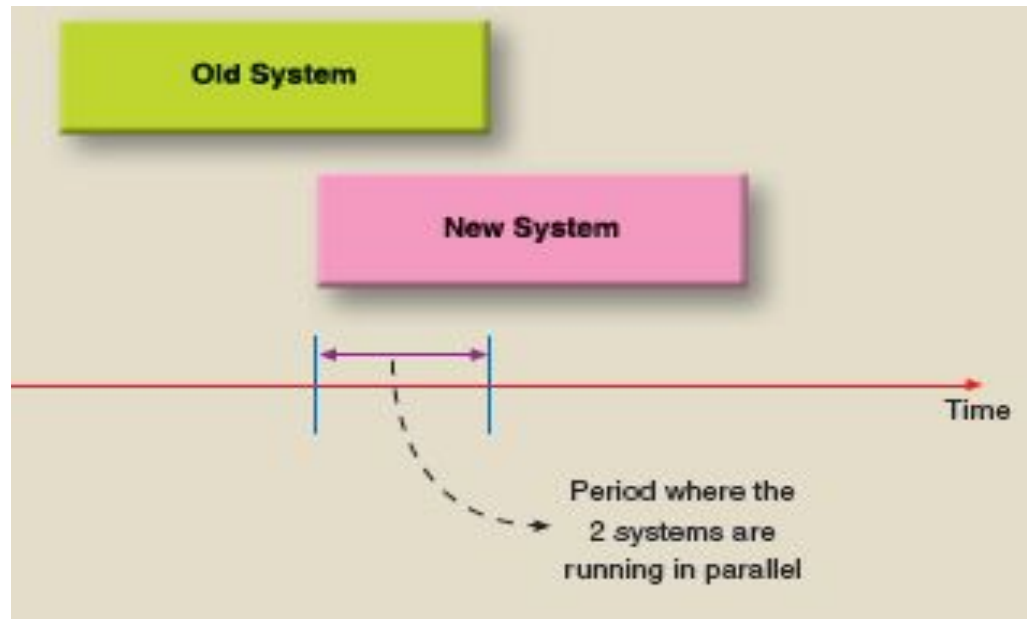
- Internal Training involving internal trainer. An internal trainer is sent to the vendor for training first.

## Reasons

- **Cost** savings, as the system is a very simple to learn.
- Training can be arranged in a **flexible** manner.
- Training is more **practical** making use of the current RFID readers and tags.
- **Assessment.** Immediate assessment can easily be carried out on the learning effectiveness.

## 4. System Conversion (System Changeover)

# Parallel Run



- **Meaning**

- **2 systems** : Running the old and the new systems simultaneously over certain period.
- **Inputs** : Using the same inputs for both systems
- **Outputs** : Outputs from both are compared and any errors are corrected.
- **Changeover** : Until the new system proves to be satisfactory where the old system is discontinued and the new one takes its place.

# Appropriate Circumstances Of Using Parallel Run

- **Large and risky systems.** Large systems have high risk of failure.
- **Lack of confidence.** Little experience in computer.
- **Sufficiency of resources.** Resource available eg. manpower, time, money.
- **Systems are comparable.** Where the old and new systems are compatible.

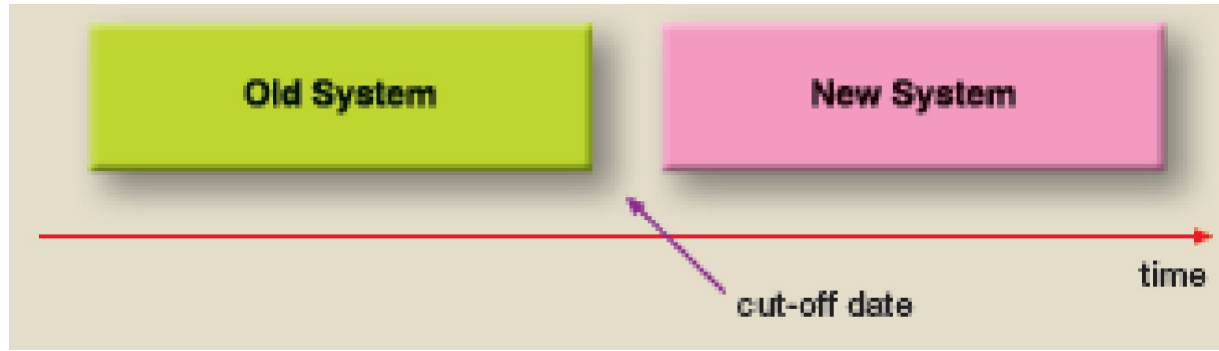
# Advantages of Parallel Run

- **Standby system.** Old system can be used as a standby when the new system failures.
- **Verification.** Old and new outputs are comparable. For checking purpose.
- **Promote user confidence.** Comparing systems helps to promote user confidence.
- **Consolidate training.** Practical is used to consolidate knowledge learnt in class.

# Disadvantages of Parallel Run

- **High cost.** Difficulties of running two systems : affect productivity, more manpower needed.
- **Cross-checking.** Difficult to cross-check errors between old and new when outputs don't match.
- **Not comparable.** Cannot be used if old and new systems are not comparable.
- **Risk of failure.** Danger of neither system being properly conducted. Tax on manpower.

# Direct Changeover ('Big Bang')



- **Meaning**

- **Short** : A short changeover method eg. days
- **Date** : Select suitable *cut-off date*, during slack periods eg. Weekends or natural breaks.
- **Changeover** : Stopping the older systems and starting the new system during cut-off date.
- **Suitable** : Adopted only when the new system bears no *resemblance* to the old and where the new system has been thoroughly *tested*.

# Appropriate Circumstances for Using Direct Changeover

- **Small system** . **Less risk of failure.**
- **Staff characteristics.** **Well trained, experienced and confident.**
- **Reliable systems.** **Where thorough testing has taken place.**
- **Not comparable.** **Where old and new systems are not comparable (same).**
- **Resources requirements.** **Resources not available.**

# Advantages of Direct Changeover

- **Flexible changeover time.** New system can be changed at any time.
- **Cheaper.** It requires the shortest time to change over.
- **Maintain morale.** A quick method of changeover, maintains morale.
- **Quick return on investment.** Users can use system to gain benefits early.

# Disadvantages of Direct Changeover

- **Risky approach.** Cannot fall back on the old system.
- **Organisation and supervision costs.** Requires well organised and supervised implementation. Increase costs.
- **Need comprehensive testing.** Requires comprehensive system tests and training. Increase costs.
- **Difficult to detect errors.** Difficult to detect errors since no comparison.

# Exercise for Students

ABC Ltd has purchased a software package for tracking its employee **attendance**. Previously, employees are required to sign on an attendance book manually. With this new system, employees need to use an RFID card and to be sensed by a device before entering the organisation.

Propose a suitable approach to **changeover**, direct or parallel run. **Justify** your proposal with 2 reasons.

# Suggested Answers

## Direct Changeover

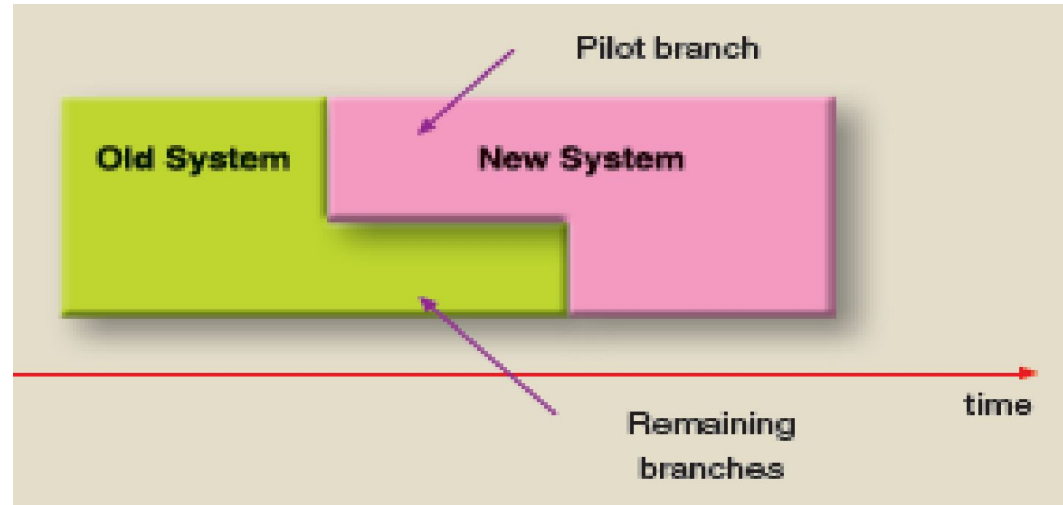
- Faster implementation
- Simple system

## Parallel

- New system. Never done that before (1st time)
- Software package is not well tested in the market place.

**Any other appropriate answers are also acceptable**

# Pilot Run



- **Meaning**

- A pilot branch or unit will be **selected**.
- **Why** ? Risk can be minimised by using one branch as a 'pilot' for the changeover.
- Upon **successful** implementation of the 'pilot' site, other branches can be changed
- Within the pilot site, it can either be a direct or a parallel run **changeover** depending on the circumstances of the case.

# Appropriate Circumstances for Pilot Run

- Used with a large system project involving **large number** of departments, units, branches or subsidiaries.
- Where the organization does not possess sufficient **resources** needed.

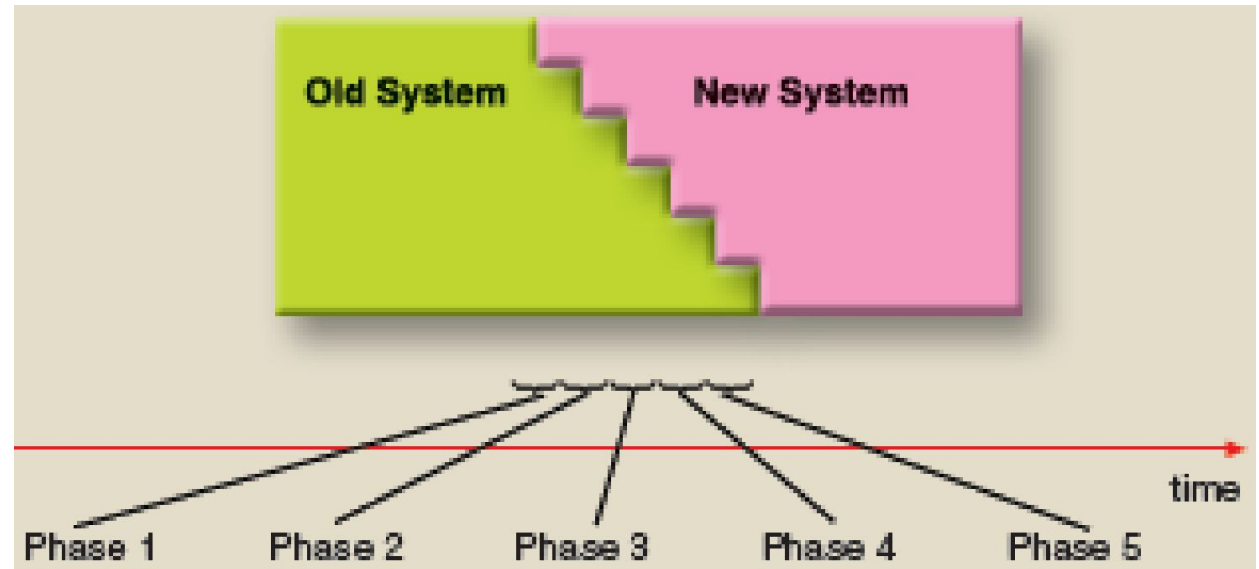
# Advantages of Pilot Run

- **Localised errors.** Errors can be identified, localised and restricted only to the pilot site, and corrected easily.
- **Cheaper.** Pilot run is cheaper than full parallel run.
- **Reduce risk of failure.** Use in large project.
- **Promote user confidence.** Successful implementation of pilot project would give confidence to others.
- **Over-stretched resources.** Pilot run would not over strain IT personnel.

# Disadvantages of Pilot Run

- **Over worked.** Users have to cope with works for both the old and new systems.
- **High investment.** Channel resources to ensure pilot project is successful.
- **Not representative.** Success of pilot would not ensure success in other branches. It does not simulate accurate business operation complexity.

# Phased or Staged Changeover



- **Meaning**

- The system is implemented in stages or time **phases**
- Involves a **series** of limited-size direct or parallel changeovers.
- The new system is being divided into a number of parts and introduced **part-by-part** (or module-by-module).
- Only when the selected part is operating satisfactorily

# Phased Changeover - Example

| Phase | Time                   | Modules                                                                                                      |
|-------|------------------------|--------------------------------------------------------------------------------------------------------------|
| 1     | Jan 2009 – Mar 2009    | Accounts Payable, Accounts Receivable                                                                        |
| 2     | April 2009 – July 2009 | Cash Management System,<br>Asset Management System                                                           |
| 3     | August 2009 – Mar 2010 | Distribution System (Warehousing,<br>Invoicing, Order Processing)                                            |
| 4     | April 2010 – Dec 2010  | Material Management<br>(Warehousing, Bills of Material,<br>Purchasing, Materials<br>Requirements Planning)   |
| 5     | Jan 2011 – June 2011   | Manufacturing Management<br>(Capacity Requirements<br>Planning, Shopfloor Control,<br>Production Scheduling) |

# Advantages of Phased Changeover

- **Less Resources Required.** Implementation is carried out in phases. Useful if there is a shortage of IT personnel or resources.
- **Reduced Risk.** Risk of errors and failures are limited only to the modules implemented.
- **User confidence.** User confidence increase on seeing successful implementation.

# Disadvantages of Phased Changeover

- **Slow process.** Implementation is slow as it is carried out phase by phase. Deprive the organization the use of the system early.
- **Staff demoralised.** May lead to staff being demoralized or loss of interest. Users' requirements may also change.
- **More expensive.** Costly since the system implementation involves a large number of separate phases.